Crime Confidence Intervals: Minneapolis Police Incidents (2015–2019)

Anjali Vemula (6076947)

2026-04-20

Packages & Data Loading

```
packages <- c("tidyverse", "ggplot2", "dplyr", "scales",
              "viridis", "knitr", "gridExtra", "ellipse")

package.check <- lapply(packages, FUN = function(x) {
  if (!require(x, character.only = TRUE, quietly = TRUE)) {
    tryCatch(
      install.packages(x, dependencies = TRUE,
                       repos = "https://cloud.r-project.org"),
      error = function(e) message(paste("Could not install:", x))
    )
  }
  tryCatch(
    library(x, character.only = TRUE),
    error = function(e) message(paste("Could not load:", x))
  )
})
```

```
Police_Incidents_2015 <- read.csv("Police_Incidents_2015.csv")
Police_Incidents_2016 <- read.csv("Police_Incidents_2016.csv")
Police_Incidents_2017 <- read.csv("Police_Incidents_2017.csv")
Police_Incidents_2018 <- read.csv("Police_Incidents_2018.csv")
# Police_Incidents_2019 <- read.csv("Police_Incidents_2019.csv")
```

Data Preparation

```
Police_Incidents_2015$Year <- 2015
Police_Incidents_2016$Year <- 2016
Police_Incidents_2017$Year <- 2017
Police_Incidents_2018$Year <- 2018
# Police_Incidents_2019$Year <- 2019
```

```r
offenses_unclean <- bind_rows(
  Police_Incidents_2015, Police_Incidents_2016, Police_Incidents_2017,
  Police_Incidents_2018
)
```

```r
offenses <- offenses_unclean %>%
  select(Year,
         offense = Offense,
         neighborhood = Neighborhood,
         date = ReportedDate,
         time = Time,
         Long, Lat) %>%
  mutate(
    date = as.Date(date, format = "%m/%d/%Y"),
    hour = as.numeric(substr(time, 1, 2)),
    time_period = case_when(
      hour >= 5  & hour < 12 ~ "Morning",
      hour >= 12 & hour < 17 ~ "Afternoon",
      hour >= 17 & hour < 21 ~ "Evening",
      TRUE ~ "Night"
    )
  ) %>%
  filter(!is.na(neighborhood),
         !is.na(offense),
         !is.na(Long), !is.na(Lat),
         Long < -93 & Long > -94,
         Lat > 44 & Lat < 46)


cat("Total cleaned records:", nrow(offenses), "\n")
```

```
## Total cleaned records: 69396
```

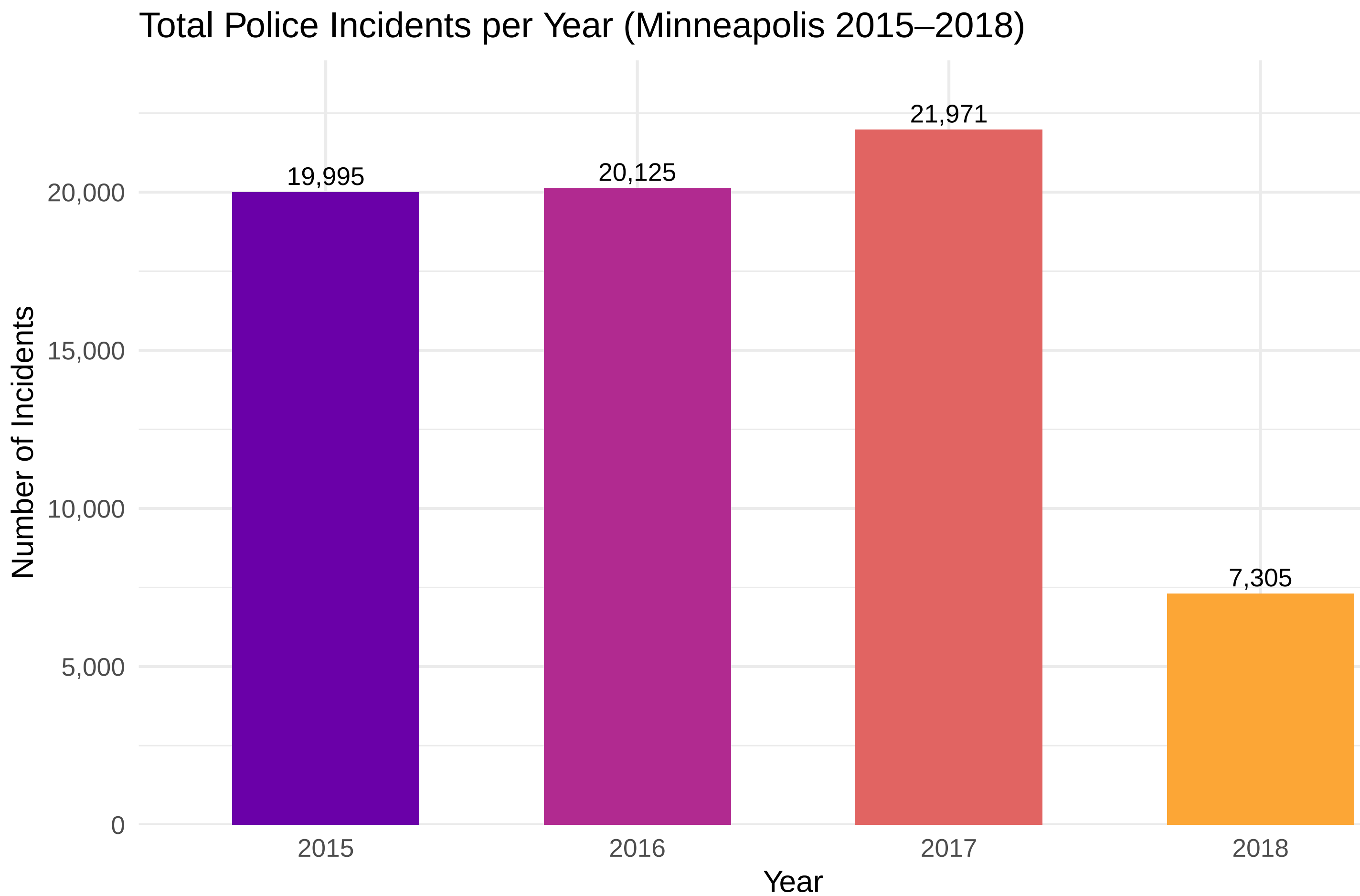
```r
cat("Years covered:", paste(sort(unique(offenses$Year)), collapse = ", "), "\n")
```

```
## Years covered: 2015, 2016, 2017, 2018
```

```r
cat("Unique neighborhoods:", n_distinct(offenses$neighborhood), "\n")
```

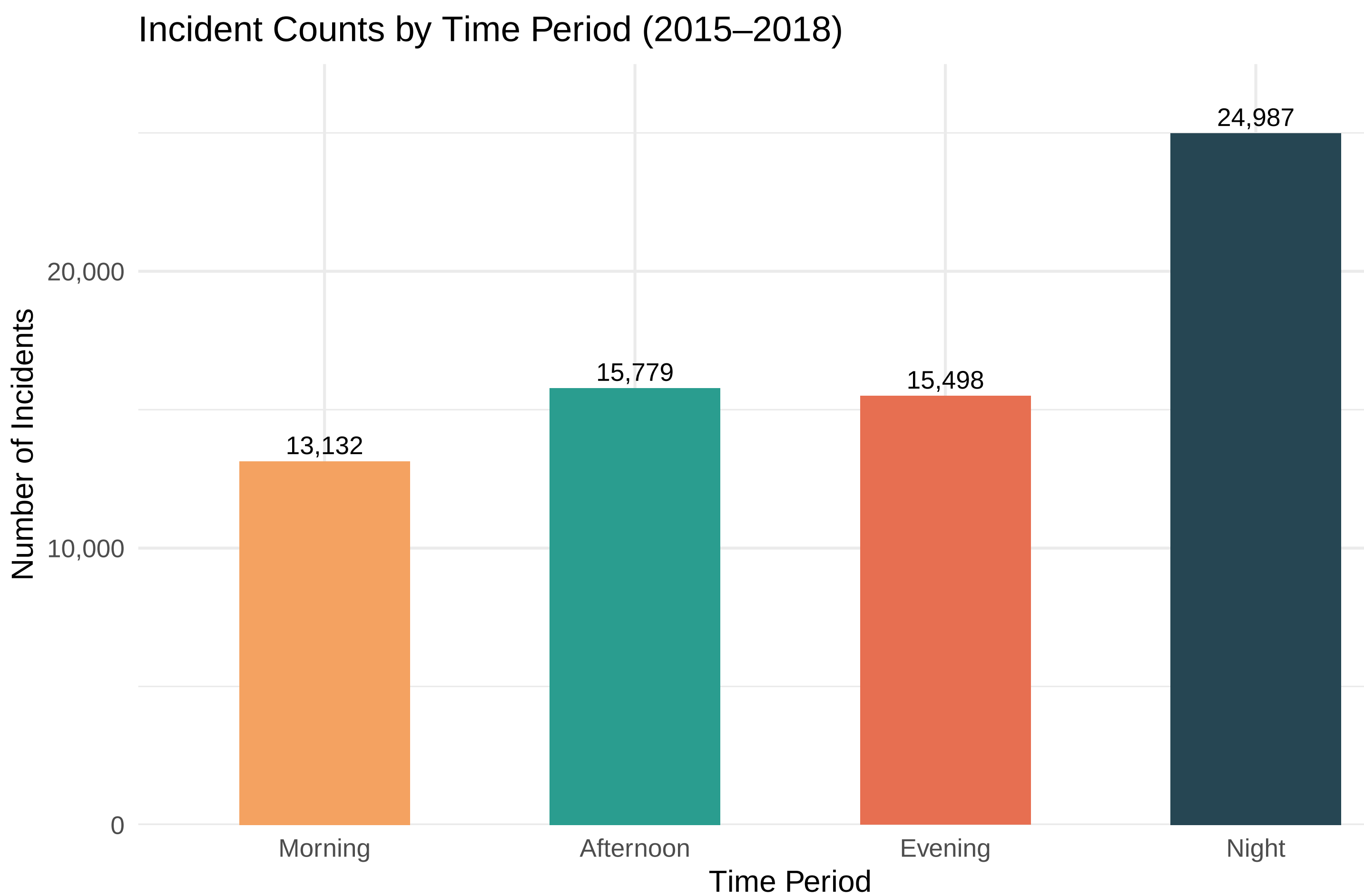
```
## Unique neighborhoods: 88
```

```r
yearly <- offenses %>% count(Year, name = "Offenses")

ggplot(yearly, aes(x = factor(Year), y = Offenses, fill = factor(Year))) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(label = scales::comma(Offenses)), vjust = -0.4, size = 3.5) +
  scale_fill_viridis_d(option = "C", begin = 0.2, end = 0.8) +
  scale_y_continuous(labels = scales::comma, expand = expansion(mult = c(0, 0.1))) +
  labs(title = "Total Police Incidents per Year (Minneapolis 2015-2018)",
       x = "Year", y = "Number of Incidents") +
  theme_minimal(base_size = 12)
```



Distribution by Time of Day

```
tp_counts <- offenses %>% count(time_period, name = "n") %>%
  mutate(time_period = factor(time_period,
                              levels = c("Morning","Afternoon","Evening","Night")))

ggplot(tp_counts, aes(x = time_period, y = n, fill = time_period)) +
  geom_col(width = 0.55, show.legend = FALSE) +
  geom_text(aes(label = scales::comma(n)), vjust = -0.4, size = 3.5) +
  scale_fill_manual(values = c("#F4A261","#2A9D8F","#E76F51","#264653")) +
  scale_y_continuous(labels = scales::comma, expand = expansion(mult = c(0, 0.1))) +
  labs(title = "Incident Counts by Time Period (2015-2018)",
       x = "Time Period", y = "Number of Incidents") +
  theme_minimal(base_size = 12)
```



Top 15 Neighborhoods

```r
top_nbhd <- offenses %>%
  count(neighborhood, name = "n") %>%
  slice_max(n, n = 15) %>%
  mutate(neighborhood = fct_reorder(neighborhood, n))

ggplot(top_nbhd, aes(x = n, y = neighborhood, fill = n)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = scales::comma(n)), hjust = -0.1, size = 3.2) +
  scale_fill_viridis_c(option = "B", begin = 0.3, end = 0.9) +
  scale_x_continuous(expand = expansion(mult = c(0, 0.15)),
                     labels = scales::comma) +
  labs(title = "Top 15 Neighborhoods by Incident Count (2015-2018)",
       x = "Total Incidents", y = NULL) +
  theme_minimal(base_size = 11)
```

```r
daily_tp <- offenses %>%
  count(date, time_period, name = "n") %>%
  pivot_wider(names_from = time_period, values_from = n, values_fill = 0) %>%
```

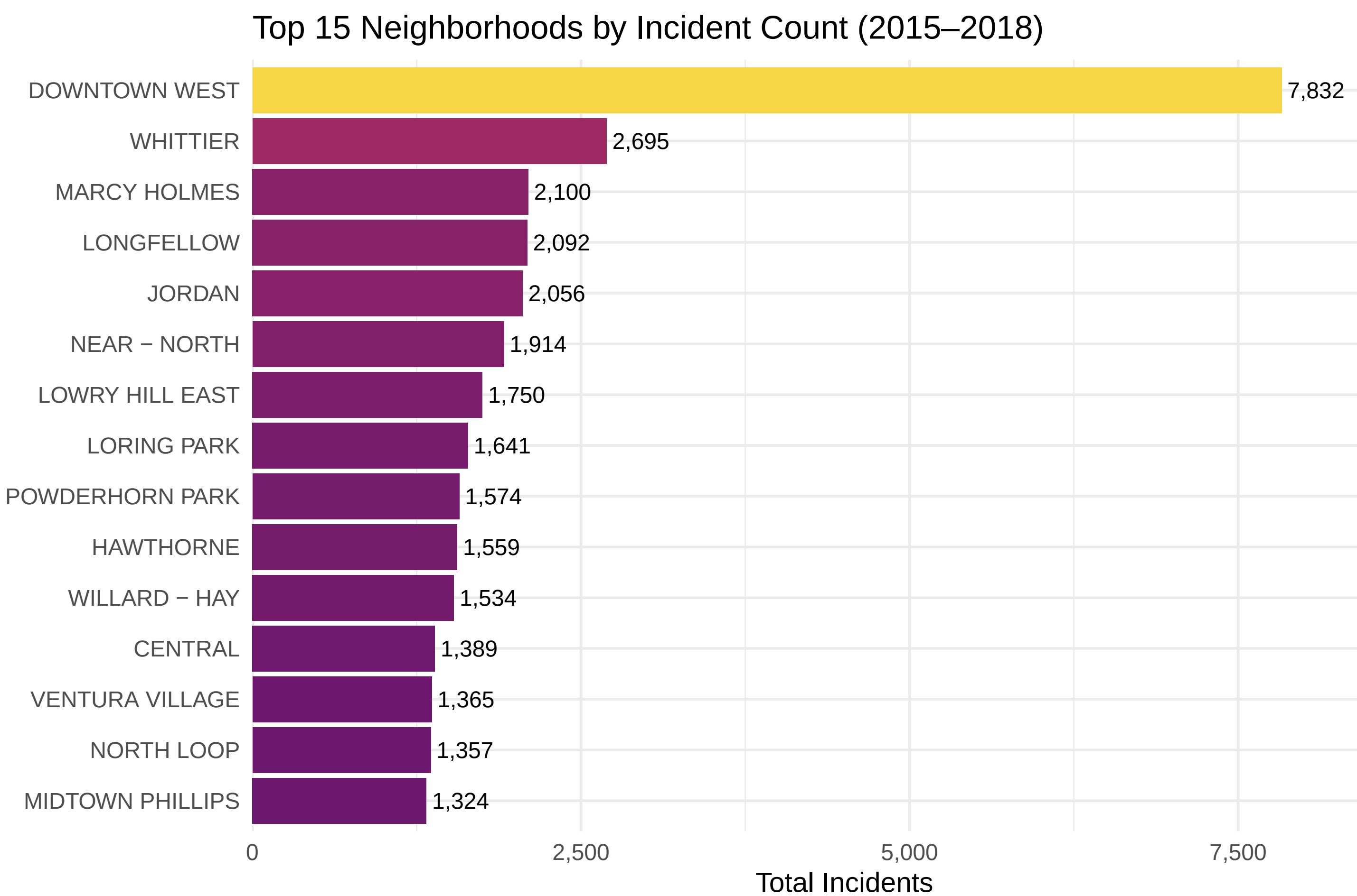


Figure 1: Downtown and Near North neighborhoods consistently report the highest incident volumes, concentrating about a third of all city offenses.

```r
  mutate(
    Morning   = if ("Morning"   %in% names(.)) Morning   else 0,
    Afternoon = if ("Afternoon" %in% names(.)) Afternoon else 0,
    Evening   = if ("Evening"   %in% names(.)) Evening   else 0,
    Night     = if ("Night"     %in% names(.)) Night     else 0
  ) %>%
  select(date, Morning, Afternoon, Evening, Night)

# Numeric matrix for multivariate analysis
X <- as.matrix(daily_tp[, c("Morning", "Afternoon", "Evening", "Night")])

n <- nrow(X)   # number of days
p <- ncol(X)   # number of variables (time periods)

x_bar <- colMeans(X)
S     <- cov(X)

cat("Sample size (days):", n, "\n")
```

```
## Sample size (days): 1251
```

```r
cat("\nSample mean vector (avg daily incidents per period):\n")
```

```
##
## Sample mean vector (avg daily incidents per period):
```

```r
round(x_bar, 3)
```

```
##   Morning Afternoon   Evening     Night
##    10.497    12.613    12.388    19.974
```

```r
summary_df <- data.frame(
  Period    = names(x_bar),
  Mean      = round(x_bar, 2),
  SD        = round(apply(X, 2, sd), 2),
  Min       = apply(X, 2, min),
  Max       = apply(X, 2, max)
)
summary_df
```

```
##              Period  Mean   SD Min Max
## Morning     Morning 10.50 4.20   1  28
## Afternoon Afternoon 12.61 4.50   0  37
## Evening     Evening 12.39 4.25   2  27
## Night         Night 19.97 6.81   3  45
```